Journal of Environmental Psychology 71 (2020) 101488

Contents lists available at ScienceDirect

Journal of Environmental Psychology

journal homepage: www.elsevier.com/locate/jep

Restoration of sustained attention following virtual nature exposure: Undeniable or unreliable?

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ARTICLE INFO

Handling Editor: Sander van der Linden

Keywords:
Attention restoration
Natural environments
Urban environments
Nature exposure
Attention
Sustained attention

ABSTRACT

Building on a body of research examining the influence of natural versus urban images on attention, the purpose of the present study was to examine attention restoration with (1) a large sample size, (2) a broader image set that was more representative of typical natural and urban environments, and (3) an increased number of task trials to increase the likelihood of more thorough attentional depletion. In order to compare the restorative effects of natural versus urban images on attention, participants completed the Sustained Attention to Response Task (SART; a measure of sustained attention) before and after they viewed either natural or urban images. When compared to the urban condition, participants in the nature condition did not demonstrate improved performance in the post-image exposure SART. Bayesian analyses also indicated support for the null hypothesis. Thus, this study provides evidence that is inconsistent with the consensus view that images of natural settings restore attention more than images of urban settings.

1. Introduction

In environmental psychology, there is vast interest in understanding how natural environments may more effectively restore depleted attentional capacities – and thus lead to improved post-exposure attentional performance – than urban environments (Berman, Jonides, & Kaplan, 2008; Berto, 2005; Kaplan & Kaplan, 1989; for a review, see; Ohly et al., 2016, or; Stevenson, Schilab, & Bentsen, 2018). To explain the potential restorative benefits of natural environments, Kaplan and colleagues (Kaplan, 1995; Kaplan & Kaplan, 1989) proposed the Attention Restoration Theory (ART), which posits that there are two distinct types of attention: 'involuntary attention', which is uncontrolled and effortless; and 'directed attention', which is limited in capacity, controlled, effortful and can be depleted. According to ART, natural environments hold more restorative potential than urban environments because they are more likely to gently, and involuntarily, guide the viewer's attention in such a way that directed attention is not required and has a chance to rest (e.g., a calmly flowing river; this situation is referred to as "soft fascination", Kaplan, 1995).

While much of the past research has examined the influence of *real-world* environments on attention, *virtual image/video* exposures have also been studied, perhaps because of the restorative potential they hold when barriers to real environments (e.g., accessibility, time, financial, and other socioeconomic factors) are present. Pioneering the efforts to test ART in a novel and virtual behavioural paradigm, Berto (2005) provided foundational support for the idea that attention performance can be restored following exposure to restorative images. In this important study, 32 participants completed two rounds of the Sustained Attention to Response Task (SART; Roberston, Manly, Andrade, Baddeley, & Yiend, 1997), separated by a the presentation of either 'restorative' images (all natural environments, $n = 16$), or 'non-restorative' images (all urban environments, $n = 16$). Images with high ratings on the *Perceived Restorativeness Scale (PRS) – Short* (Berto, 2001; Korpela & Hartig, 1996; Peron, Berto, & Purcell, 2002) were selected for the restorative category, while images with low ratings were selected for the non-restorative category. In the SART, the digits from 1 to 9 are presented one at a time at a constant rate on the computer screen. Participants are required to make key-press responses to non-target stimuli (e.g., 1, 2, 4-9) and withhold the key-press response to a target stimulus (e.g., 3). Berto (2005) found that, compared to those who viewed the lower perceived restoration images, participants who viewed higher perceived restoration images made significantly more correct withholds on target trials and had significantly faster response times during the post-image-exposure SART in comparison to the

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https://doi.org/10.1016/j.jenvp.2020.101488
Received 15 June 2020; Received in revised form 11 August 2020; Accepted 11 August 2020
Available online 15 August 2020
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pre-image-exposure SART. Based on these results, Berto (2005) concluded that the restorative image set was indeed restorative, as indexed by performance on an attention task, while the non-restorative image set was not. Importantly, Berto (2005) also noted that the present research "underscore(s) ART's premise that natural environments are more restorative than urban environments", (Berto, 2005, p. 258).

In the present investigation, our goal was to more directly test the conclusion that natural environments restore attention more effectively than do urban environments. As such, the present study is a conceptual replication focusing on one conclusion drawn from Berto's work. We chose to focus on this conclusion because Berto's (2005) findings are commonly cited as showing that nature environments confer greater attentional restoration than urban environments (e.g., Abraham, Sommerhalder, & Abel, 2010; Atchley, Strayer, & Atchley, 2012; Berman et al., 2008; Bratman, Daily, Levy, & Gross, 2015; Bratman, Hamilton, & Daily, 2012; Carrus, Passiatore, Pirchio, & Scopelliti, 2015; Kaplan & Berman, 2010; Sonnentag, Venz, & Casper, 2017; van den Berg, Maas, Verheig, & Groenewgen, 2010). Importantly, this common interpretation of Berto's (2005) findings might not be necessitated by the data presented by Berto (2005) because the images used in Berto's study were selected based on predicted *restorativeness* (high vs. low), and not exclusively on *environment type* (nature vs. urban; though this distinction was also present within the stimulus set). We hasten to add however, that Berto (2005) might have been primarily interested in the influence of 'restorativeness' on attention, and so our study is not a direct replication of Berto's work.

Studies that have followed up on Berto's (2005) findings have several critical limitations. First, several attempted replications have continued to use stimuli confounded by restorativeness (e.g., Craig, 2016; Craig & Klein, 2012; Craig, Klein, Menon, & Rinaldo, 2015; Dumke, 2014; Nguyen, Nelson, & Klein, 2018). Second, in several studies, researchers used only one stimulus in each of the urban and nature conditions (e.g., Lee, Williams, Sargent, Williams, & Johnson, 2015; Sanguinetti, 2009; Valtchanov, 2010), thus substantively limiting their generalizability. Finally, we note that the extant replications of Berto's (2005) findings have yielded mixed results (e.g., Craig, 2016; Dumke, 2014; Nguyen, Nelson, & Klein, 2018; Sanguinetti, 2009). The inconsistency in the results across these studies may be in part due to the underwhelming sample sizes used: 16 participants per group in Berto's study and 25 participants or less per group in a number of publications (e.g., Craig, 2016; Craig et al., 2015; Craig & Klein, 2012; Dumke, 2014; Nguyen, Nelson, & Klein, 2018; Sanguinetti, 2009; Valtchanov, 2010). Thus, it is clear that what is necessary is a well-designed and well-powered conceptual replication of the commonly cited conclusion from Berto (2005), namely that images of natural environments are more restorative than urban environments.

1.1. The present study

Given the abovementioned concerns, the purpose of the current study is threefold. First, we aimed to more directly test the conclusion drawn by Berto (2005) as it is commonly cited in the literature—that exposure to natural environments is more attentionally restorative than exposure to urban environments. To gauge this, we collected participants' ratings of how *typical* various images were of natural and urban settings, rather than on the basis of restorativeness. Second, we aimed to test Berto's (2005) conclusion with a larger sample that would be sufficient to detect a small effect size. We determined that 80 participants per group (160 participants total) would be required to detect a fairly small effect size ($d = 0.30$), with a power of .80 and a two-tailed alpha value of 0.05. Finally, in our study, we increased the number of trials in an attempt to more thoroughly deplete attention during the pre-exposure SART, and to increase the likelihood we would observe any potential attention restoration during the post-exposure SART. Since sustained attention performance declines as a function of time-on-task (a.k.a., the 'vigilance decrement'; Head, Russell, Dorahy, Neumann, & Helton, 2012; Mackworth, 1948), we predicted that attention depletion would be the greatest at the end of the pre-exposure SART trials, and that attention restoration would be the strongest at the beginning of the post-exposure SART trials.

2. Materials and methods

2.1. Participants

Two hundred students (*mean age* = 19.74, *SD* = 1.87) from the University of Waterloo participated in this study in exchange for partial course credit. Two participants did not complete the experiment due to technical difficulties and thus could not be included in the analyses. Thirteen students were excluded from the analysis as they were tested during preliminary sessions of the experiment used to train the research assistant, and five participants were excluded due to deviations in the experimental protocol (e.g., the presence of visual/auditory distractions due to other participants; participants completing the SART in other experiments just prior to the present experiment).[1] Thus, a total of 180 participants were included in the analysis (123 female, 54 male, 2 unknown). Slight overshooting of our planned sample was due to participants being run in groups and the anticipation of 'no-shows'. Participants were randomly assigned to either the nature or the urban condition and were run in group sizes ranging from 1 to 4 participants.

2.2. Ethics statement

This study was reviewed and approved for ethic clearance through the University of Waterloo Research Ethics Committee (ORE #40442).

2.3. Materials

2.3.1. Image selection

One hundred images were selected from various free desktop background websites. Fifty images contained mostly natural settings and 50 images contained mostly urban settings. The goal was to sample a wide selection of images that varied on a spectrum from natural to urban environments. No settings contained people. Image selection was based on the types of environments that Berto (2005) included in the original picture set: lakes, rivers, seas, hills, woods, orchards, forests, city riversides, city streets, industrial zones, housing, porches, urban areas, and skyscrapers.

Prior to the commencement of the experiment proper, a pre-scaling study was conducted to determine which subset of images from the initial set of 100 images would be included in the experiment proper. In this pre-scaling study, 100 participants were recruited from the Amazon Mechanical Turk (www.mturk.com). These participants were compensated $3.50 USD for a 30-min study. Participants viewed each image for 6 s. After viewing each image, participants were asked to respond to two slider scales. The first slider scale read: "That is a place that matches my idea of a typical natural environment", and the second slider read: "That is a place that matches my idea of a typical urban environment"; both sliders were anchored by, "(0) Strongly Disagree" and "(100) Strongly Agree".

The ratings of the images from the pre-scaling study were then used

[1] These exclusions were not pre-registered. For full transparency, the data were analyzed both with and without these data points; however, the two participants who experienced technical difficulties (noted in Section 2.1) were still excluded for all analyses as they did not have data for the post-exposure SART and thus could not be included. Importantly, both sets of analyses led to the same general conclusions. Analyses in the manuscript are reported excluding the abovementioned participants. However, analyses including these participants are reported in the footnotes in the relevant sections of the Results and a full report of these analyses can be found our Supplemental Materials.

to select which images would make up the stimuli sets for the urban and natural conditions in the present study. The ratings for each image was averaged across participants. Difference scores were calculated for each image by subtracting the average urban typicality values from the average nature typicality values. Then, the twenty-five images with the largest positive difference scores were selected as the images for the nature condition, and the twenty-five images with the largest negative difference scores were selected as the images for the urban condition. Once the images were selected, they were balanced on contrast and luminance using Matlab programming (McCrackin & Itier, 2018; Willenbockel et al., 2010).

2.3.2. Sustained attention to Response Task (SART)

The SART (Robertson, Manly, Andrade, Baddeley, & Yiend, 1997) is a computer task in which participants are presented a series of digits (1-9) one at a time and must actively monitor for infrequent targets. Participants were instructed to press the space bar for each "go" non-target digit (*1, 2, 4-9*), and withhold responses to the "no-go" target digit, (*3*). For each trial, a single digit from 1 to 9 was randomly selected without replacement to appear for 250 ms in the center of the screen, followed by a "+" mask presented for 900 ms, resulting in a total trial duration of 1150 ms. The critical no-go target digit was presented pseudo-randomly over the duration of the task. The SART was comprised of 70 blocks of nine trials each, for a total of 630 trials, resulting in a total task length of 12 min and 4.5 s. All digits were presented in white on a black background, using the Courier New typeface in one of five randomly selected sizes: 120, 100, 94, 72, and 48 points.

The primary measure of interest obtained from the SART was commission errors, defined as instances when participants failed to withhold a response to a no-go trial and an important index of failures in sustained attention (Robertson et al., 1997). Secondary measures of attention lapses included go-trial reaction times (RTs), defined as the mean response latencies for all of the go-trials in which a response was made; and omission errors, defined as instances when participants failed to respond to a go-trial (Cheyne, Solman, Carriere, & Smilek, 2009; Robertson et al., 1997).

2.3.3. Image viewing task

Fifty images (25 per condition) selected from the pre-scaling study were presented in colour with dimensions of 800x800px in the center of the screen. The rest of the screen surrounding the images was black. Each image was presented for 15 s, resulting in a total image-viewing period of 6 min and 15 s.

2.4. Procedure

Participants were tested in groups, ranging in size from one to four. Prior to the experimental session, participants were randomly assigned to the nature or urban condition, which determined the type of images that were presented. Participants provided consent and were then given verbal (by the experimenter) and written instructions in the computer program regarding the experimental task. Following the instructions, participants were asked to place headphones over their ears to prevent distraction from other sounds (e.g., key presses) in the room. Participants completed the practice trials, and if they misunderstood the goal of the task, the experimenter corrected them. Participants then began the experiment proper and the experimenter left the room. Participants completed the 630 trials in the pre-exposure SART, then they viewed 25 images in their respective condition (i.e., either nature or urban), and finally they completed 630 trials in the post-exposure SART. The experiment was approximately 50 min in length.

3. Results

3.1. Data exclusion and analysis overview

Data was first analyzed for violations of normality. Initial descriptive statistics revealed that, while skew and kurtosis values were normal for response times and commission errors, the outputs for omission errors were not within the acceptable skew and kurtosis range ($|skew| < 3.0$ and $|kurtosis| < 10$, as per the recommendations of Kline, 1998). This was due to a few outliers in the high end of the omission distribution. To correct for this, participants with an average omission error rate of 3 standard deviations or greater from the mean was excluded listwise from each condition pre- and post-image exposure (as per of the protocol of Seli, Jonker, Solman, Cheyne, & Smilek, 2013). This was done as higher omission rates may be indicative of participants who failed to understand the instructions or willfully decided not to follow instructions. After removing those participants, the skew and kurtosis values for the omissions of each group fell within the normal range (Kline, 1998). The final sample consisted of 171 participants, with 84 participants in the nature condition, and 85 participants in the urban condition.

The data were analyzed in two ways. The first analysis examined each of the SART dependent variables (commission errors, RTs, and omission errors) based on data from all trials of the pre- and post-exposure SARTs. The second analysis included only the last 243 trials of the pre-exposure SART and the first 243 trials of the post-exposure SART, in an attempt to more directly mimic Berto's (2005) analysis. In addition, analyses are reported with omission outliers both removed and included.

To determine whether the data provided evidence for the null hypothesis (i.e., that image type does not influence SART performance), we conducted a Bayes analysis. The Bayes factor allowed us to assess the evidence in favour of the null hypothesis (H_0) and the alternative hypothesis (H_1) given a distribution for the prior probability of these hypotheses. A BF_{01} is the ratio of the likelihood of the data under the null hypothesis compared to the alternative hypothesis. We interpreted the Bayes factors based on the recommendations of Jeffreys (1961).

3.2. Analyses of all SART trials

The data for each performance metric of the full-length pre- and post-image exposure SARTs were submitted to separate 2×2 mixed ANOVAs with Image Type (Nature vs. Urban) as a between participant variable and SART Block (Pre-exposure vs. Post-exposure) as a within-participant variable. The means (and confidence intervals) of each of the SART dependent variables for each of the pre-and post-exposure SARTs, averaged across participants in each of the Image Type groups, are shown in Table 1. Boxplots for all three dependent variables from all SART trials are portrayed in Fig. 1.

Table 1
Descriptive statistics for all SART trials.

Image Type	SART Block	N	Commission Errors	Reaction Time	Omission Errors
Urban	Pre-Exposure	86	.47 [.43, .52]	381 [361, 401]	.02 [.02, .03]
	Post-Exposure	86	.49 [.44, .54]	382 [362, 403]	.04 [.03, .06]
Nature	Pre-Exposure	85	.44 [.40, .48]	390 [371, 409]	.02 [.01, .02]
	Post-Exposure	85	.45 [.40, .50]	386 [366, 405]	.03 [.02, .03]

Note. Means are reported for each variable and condition. Square brackets denote upper and lower limits of 95% confidence intervals.

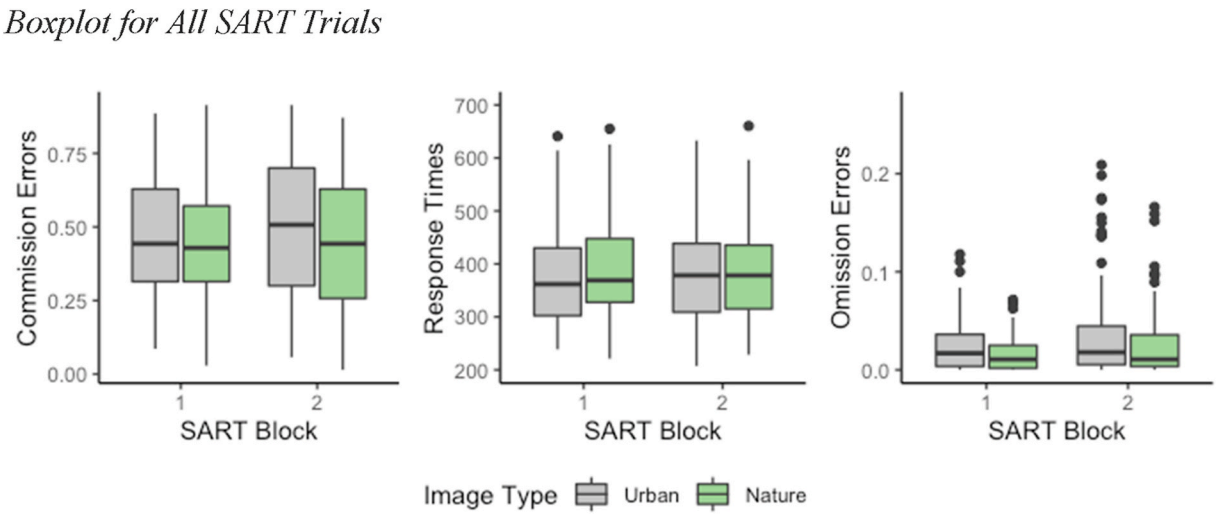


Fig. 1. Boxplot for all SART trials.
Note. This figure contains boxplots representing data from each measure when all SART trials are included. From left to right, the graphs show data from commission errors, response times, and omission errors. Data from the urban condition are portrayed in grey, while data from the nature condition are portrayed in green.

Table 2
Descriptive statistics for reduced SART trials.

mage Type	SART Block	N	Commission Errors	Reaction Time	Omission Errors
Urban	Pre-Exposure	86	.52 [.46, .57]	384 [361, 406]	.03 [.02, .04]
	Post-Exposure	86	.47 [.41, .52]	381 [359, 403]	.03 [.02, .05]
Nature	Pre-Exposure	85	.49 [.44, .54]	387 [365, 409]	.02 [.02, .03]
	Post-Exposure	85	.42 [.37, .48]	383 [362, 404]	.02 [.01, .03]

Note. Means are reported for each variable and condition. Square brackets denote upper and lower limits of 95% confidence intervals.

3.2.1. Commission errors

When examining the *commission errors*, there were no significant main effects of either SART Block, $F(1, 169) = 1.96$, $p = .163$ $\eta^2 = 0.01$, or Image Type, $F(1, 169) = 1.62$, $p = .205$, $\eta^2 < 0.01$. There was also no significant interaction between SART Block and Image Type, $F(1, 169) = 0.59$, $p = .445$, $\eta^2 < 0.01$. The Bayes factor also provided substantial evidence that a model excluding the interaction better approximates the data than a model with the interaction, $BF_{01} = 5.04$. Without excluding outliers, all tests remain non-significant (all $F(1, 178) < 1.49$, all $p > .224$, and evidence against favoring a model with the interaction was again substantial $BF_{01} = 6.03$).

3.2.2. Reaction times

When examining the *reaction times*, there was no main effect of SART Block, $F(1, 169) = 0.21$, $p = .646$, $\eta^2 < 0.01$, no main effect of Image Type, $F(1, 169) = 0.21$, $p = .646$, $\eta^2 < 0.01$, nor a significant interaction between SART Block and Image Type, $F(1, 169) = 0.53$, $p = .467$, $\eta^2 < 0.01$. The Bayes factor provided substantial evidence against the model with the interaction, $BF_{01} = 5.02$. With no exclusions, all tests remain non-significant (all $F(1, 178) < 0.95$, all $p > .330$, and evidence against the inclusion of the interaction remained consistent, $BF_{01} = 4.34$).

3.2.3. Omission errors

When examining the *omission errors*, there was a significant main effect of SART Block, $F(1, 169) = 25.53$, $p < .001$, $\eta^2 = 0.13$, such that participants made more omission errors in the post-exposure SART than pre-exposure SART. In addition, there was a significant main effect of Image Type, $F(1, 169) = 5.99$, $p = .015$, $\eta^2 = 0.03$, whereby participants in the Urban condition made more omission errors than those in the Nature condition. However, the interaction between SART Block and Image Type did not reach significance, $F(1, 169) = 2.88$, $p = .092$, $\eta^2 = 0.02$. There was anecdotal evidence against the inclusion of the interaction, with a Bayes factor of $BF_{01} = 1.78$. With no exclusions, the main effects of SART Block and Image Type remained significant ($F(1, 178) = 17.62$, $p < .001$; $F(1, 178) = 4.05$, $p = .046$, respectively), while the interaction remained non-significant ($F(1, 178) = 1.41$, $p = .236$). However, with no exclusions, the Bayes factor now provided substantial evidence against the inclusion of the interaction, $BF_{01} = 3.36$[2].

3.3. Analyses of reduced SART trials

To examine SART performance in the number of trials that better matched the number of trials used in Berto (2005)—which only included 240 trials per SART—each performance metric was submitted to separate 2 × 2 mixed ANOVAs conducted with only the last 243 trials of the pre-exposure SART and the first 243 trials of the post-exposure SART. As with the previous analysis, Image Type (Nature vs. Urban) was included as a between participant variable and SART Block (pre-exposure vs. post-exposure) was included as a within-participant variable. Descriptive statistics for SART performance metrics corresponding to the reduced SART trials are included in Table 2. Boxplots for all three dependent variables from these reduced SART trials are portrayed in Fig. 2.

[2] When the excluded data points (see Section 2.1) were included in the full SART trial analyses (also including outlying omission errors), all null results continued to hold (see Section 3.2.). In addition, while the main effect of Image Type was significant for omission errors in the initial analyses with the aforementioned exclusions (see Section 3.2.3.), there was no main effect of Image Type when the excluded data points were included. The main effect of SART Block remained significant. Importantly, all interactions remained null when the previously excluded data points were included in the full SART analysis and Bayes factors also provided evidence against the inclusion of an interaction model of the data.

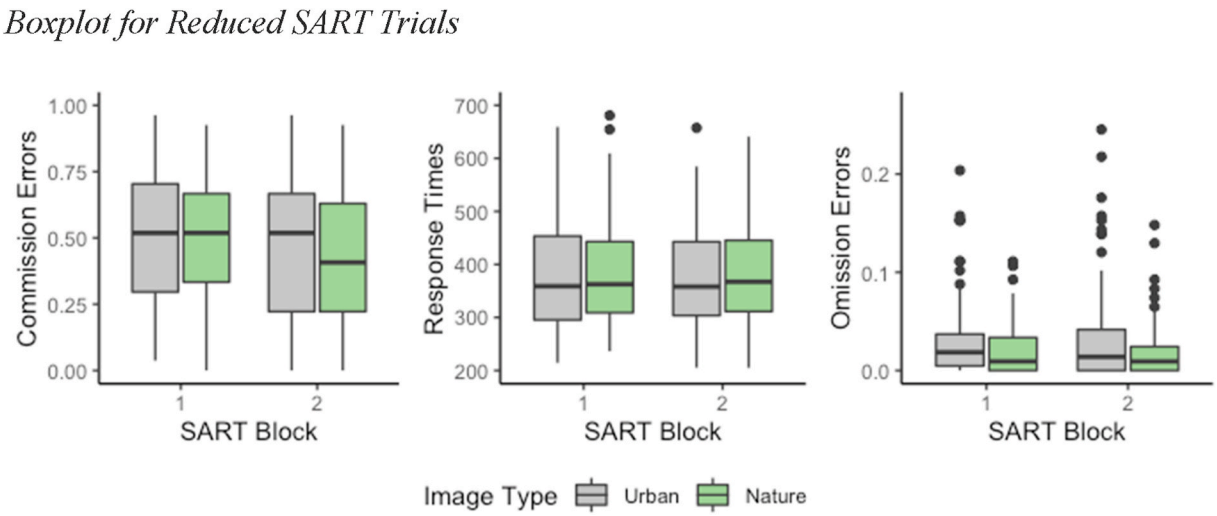


Fig. 2. Boxplot for reduced SART trials.
Note. This figure contains boxplots representing data from each measure when a reduced number of SART trials are included. From left to right, the graphs show data from commission errors, response times, and omission errors. Data from the urban condition are portrayed in grey, while data from the nature condition are portrayed in green.

3.3.1. Commission errors

When examining the *commission errors* of the last 243 trials of the pre-exposure SART and the first 243 trials of the post-exposure SART, there was a main effect of SART Block, $F(1, 167) = 22.11$, $p < .001$, $\eta^2 = 0.12$, such that less commission errors were made (regardless of Image Type) during the post-exposure SART compared to the pre-exposure SART. However, there was no significant main effect of Image Type, $F(1, 167) = 1.15$, $p = .286$, $\eta^2 < 0.01$, and no significant interaction between SART Block and Image Type, $F(1, 167) = 0.48$, $p = .489$, $\eta^2 < 0.01$. The Bayes factor was also calculated and it provided substantial evidence that the model without the interaction was better than the model with an interaction, $BF_{01} = 5.23$. With no exclusions, the main effect of SART Block remained significant, $F(1, 178) = 21.32$, $p < .001$), while the main effect of Image type and the interaction remained non-significant (both $F(1, 178) < 0.93$, both $p > .335$). The Bayes factor continued to provide substantial evidence against the inclusion of the interaction, $BF_{01} = 5.49$.

3.3.2. Reaction times

When comparing the *reaction times* of the last 243 trials of the pre-exposure SART and the first 243 trials of the post-exposure SART, there was no main effect of SART Block, $F(1, 167) = 0.94$, $p = .334$, $\eta^2 < 0.01$, no main effect of Image Type, $F(1, 167) = 0.04$, $p = .842$, $\eta^2 < 0.01$, nor an interaction between SART Block and Image Type, $F(1, 167) = 0.05$, $p = .818$, $\eta^2 < 0.01$. The Bayes factor indicated substantial evidence against the inclusion of the interaction in a model of the data, $BF_{01} = 6.39$. With no exclusions, all tests remain non-significant all ($F(1, 178) < 1.93$, all $p > .167$), and the Bayes factor continued to provide substantial evidence against the inclusion of the interaction, $BF_{01} = 5.63$.

3.3.3. Omission errors

When comparing the *omission errors* of the last 240 trials of the pre-exposure SART and the first 240 trials of the post-exposure SART, there was a significant main effect of Image Type, $F(1, 167) = 5.35$, $p = .022$, $\eta^2 = 0.03$, such that participants in the nature condition made fewer omission errors (regardless of exposure period) than those in the urban condition. However, there was not a significant main effect of SART Block, $F(1, 167) = 0.14$, $p = .708$, $\eta^2 < 0.01$, and there was not a significant interaction between SART Block and Image Type, $F(1, 167) = 0.57$, $p = .451$, $\eta^2 = 0.01$. Again, the Bayes factor provided evidence favoring a model of the data without an interaction, $BF_{01} = 4.75$. With no exclusions, the main effect of Image Type remained significant, $F(1, 178) = 4.21$, $p = .042$, while both the main effect of SART Block and the interaction remained non-significant (both $F(1, 178) < 1.96$, both $p > .163$). The Bayes factor continued to provide substantial evidence against the inclusion of the interaction, $BF_{01} = 5.81$[3].

4. Discussion

In a well-powered study, we found no evidence for the claim that post-exposure attention performance was better following exposure to natural, rather than, urban images. In fact, Bayesian analyses suggested strong evidence *for the null hypotheses*. There are several possible reasons why the present study did not find the same pattern reported by Berto (2005). First, our approach differed from Berto's such that our stimuli were selected based on ratings of typicality rather than restorative potential (to avoid confounding environment type with predicted restorativeness level). Second, each iteration of the SART in our study consisted of 630 trials, which is substantially more the 240 trials included in Berto's (2005) original study. However, this concern is alleviated by the fact that we did not find any difference in performance across conditions when only the first 243 trials of the post-exposure SART were compared to the last 243 trials of the pre-exposure SART (matching Berto's analysis). Finally, we should consider the possibility that the results reported by Berto (2005) might be unreliable owing to the relatively small sample used in the study.

It is important to note that while we failed to find an influence of environment (nature vs. urban) on performance in the SART following virtual exposure, this does not mean that null findings will also occur in

[3] When the previously noted data exclusions (see Section 2.1) were included in the reduced SART trial analyses (which also included outlying omission errors), all null results continued to hold (see Section 3.3.). An additional null result occurred such that, while the main effect of Image Type was significant for omission errors when the aforementioned data was excluded (see Section 3.3.3.), there was no main effect of Image Type when the those data points were included. The main effect of SART Block held for commission errors. Importantly, all interactions remained null when exclusions were included in the full SART analysis and Bayes factors also provided evidence against the inclusion of an interaction model of the data.

well-powered examinations using other types of attention tasks or other methods of nature exposure (i.e., real-world exposure). Stevenson, Schilhab, and Bentsen (2018) reviewed several studies with tasks that may hold greater potential for capturing subtle improvements in performance following natural vs. urban exposure (i.e., the Stroop Task and the Trail Making Task – version B); however, the authors noted that many of these studies were underpowered, had only small to moderate effect sizes, and demonstrated no attentional improvements when the groups within the studies started from equivalent baseline measures (Stevenson et al., 2018). With the publication of this replication, we hope to encourage further well-powered replications to provide more decisive evidence regarding the possibility of attention restoration following nature exposure across attention tasks and experimental contexts.

CRediT authorship contribution statement

Lydia J. Hicks: Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft, Writing - review & editing. **Alyssa C. Smith:** Formal analysis, Writing - original draft, Writing - review & editing. **Brandon C.W. Ralph:** Software, Writing - review & editing. **Daniel Smilek:** Supervision, Conceptualization, Methodology, Writing - original draft, Writing - review & editing, Funding acquisition.

Declaration of competing interest

This research was supported by a Natural Science and Engineering Research Council of Canada (NSERC) Discovery Grant to Daniel Smilek (2019–04071).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvp.2020.101488.

Author note

Open Access: Data, materials, and the program used in the present experiment can be found at https://osf.io/p42wm/.

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